

Watch Your Language: Investigating Content Moderation with LLMs

The motivation of the paper is to explore the performance of LLMs in text based content moderation, particularly for rule based community content moderation and toxic content detection. The authors observe that GPT-3.5 is effective at rule based content, achieving a median accuracy and precision of 64% and 83% respectively. However, the authors note that recent increase in model size only adds marginal benefit to toxicity detection.

The authors evaluate OpenAI’s ChatGPT (3, 3.5 and 4), Google’s Gemini and Meta’s LLAMA. For rule based community moderation tasks, the authors instantiate 95 subreddit specific LLMs by prompting GPT-3.5 with corresponding subreddit rules. The LLMs are then prompted with real comments sourced from each community. For toxicity detection, the authors construct a dataset of 10K comments and compare the aforementioned models’ performance. The authors observe that the GPT-3.5 model is more likely to moderate based on restrictive rules compared to prescriptive rules. The authors also note stark differences when underlying models change over time, leading to unpredictable results. For toxicity detection, LLMs outperform existing toxicity classifiers with GPT 3.5 achieving most balanced performance, albeit missing implicit toxicity.

For moderation decisions, the authors collected 500 comments from each subreddit that were removed by the moderators and another 500 which weren’t, to create a balanced dataset. GPT 3.5 performance varies across subreddits, showing high accuracy for some while performing worse than a coin toss for others. However, the authors do not note that the subreddits where GPT 3.5 performed bad were centered around discussion with experts and the moderation requires significant world context, which is a limitation of their setup. Incorporating reddit’s site-wide rules and chain of thought (CoT - prompting the model to think step by step), the performance improved considerably for subreddits requiring world context. The authors also observe higher false positive rate as compared to false negatives, specially when moderating NSFW subreddits. Over time performance also varied significantly, with updates favoring reducing false negatives over false positives, and the model showing a decrease in median recall. GPT-3.5 also fails to express all rules uniformly when prompted to indicate which rules are violated when it flags a comment for moderation. Similar spread is seen for real time moderation. Across LLMs, GPT 3.5 favors precision while Gemini pro favors recall.

For toxicity detection, the models were prompted with the definition of toxic content used by Google’s Jigsaw and were asked to rate the toxicity on a scale of 1-10. The LLMs outperform Perspective API across all metrics. Changing prompts and applying chain of thought does not have an impact on the performance. The authors also observe that varying classification thresholds meaningfully impacts precision. The toxicity ratings also align with human raters. LLMs also tend to make mistakes as they tend to trigger on poor language and stereotypes, despite the content being benign. LLMs also miss toxic content that does not contain any direct threat or insult to another individual. LLMs also did not flag some content that they deemed as sarcasm or humor, despite being flagged by humans. LLAMA 2, by comparison, is difficult to work with, producing invalid responses 56% of the time, but shows decent performance for valid responses.

Lastly authors present some discussion points and recommendations. First recommendation for LLM providers is to have more transparency in LLM decisions, as performance appears to change over time. Secondly, there room for improvement in rule-based moderation with behavior and context. LLM performance is also only marginally better than existing services for toxicity detection.

The work is interesting but the timing might be a little late. Reddit made changes to their public API pricing about 9 months ago, making it exorbitantly expensive (the paper mentions high prices for accessing Reddit’s API). Most reddit communities now lack moderation bots. It would have been interesting to see how these rule enforcing bots fare in comparison to LLMs (marked usually by the bot’s response to the removed comment detailing the reason for removal). The study does not make it evident whether the deleted comments were removed because the user account was deleted (by themselves

or due to other reddit rule violation) or if the comments were deleted by community moderators or by reddit itself.

Language Models as Fact Checkers?

As the title suggests, the motivation of the paper is to explore the potential of LLMs as fact checkers, without any external knowledge or explicit retrieval components. Prior work has dealt with open-domain question answering ability of LLMs but haven't taken fact checking into account. Although the work does not outperform methods with explicit knowledge bases, the authors argue that their method is viable and has room for further exploration. Recent works have shown that LLMs contain a large amount of world knowledge, mined from Wikipedia and other sources, but focus particularly on open-domain question answering, rather than fact-checking. Their proposed approach replaces document retriever and evidence selector models in traditional fact checking with a single LLM, making the procedure simpler, requiring fewer resources and computation, and removing the need to maintain an explicit knowledge base external to the LM.

The authors conducted a human review study, where participants were given a claim from the fact extraction and verification (FEVER) shared task, which uses modified or unmodified claims extracted from Wikipedia, and were asked to validate the claim using a BERT language model. They found that the users had reasonable in determining claim validity. Using the results of the study, the authors designed an end to end neural approach for utilizing BERT as a fact checker. The experimental steps involved masking a token from a claim and getting the top-1 predicted token from the LM. If the predicted token matches the masked token, the claim is supported, otherwise it is refuted. Average accuracy was recorded to be 55%. They repeated the experiment by always masking the last token and were able to obtain 51% precision and 59% F1 scores.

For the end-to-end fact checking model, the authors use automatic masking using Named Entity Recognition model to mask the last named entity in the claim and use a textual entailment model to validate LM predictions. Their experimental results show that their model outperforms BERT on all metrics and achieves a comparable accuracy score to FEVER baseline without the need for explicit document retrieval and evidence selection. These results suggest that language models perform on par with existing fact checking methods. The authors also provide a detailed analysis of error cases to facilitate future work. One common form of error is that entity type of token prediction is biased towards the sentence structure of the original text. For example, in Wikipedia, birth years are always preceded by the person's name, so sentences like "[Subject] was born in" always result in the model predicting the birth location, rather than the year. Likewise, general prefixes like "is a" are hard to make predictions against. Short claims (less than 5 tokens) also affect the performance.

This paper presented a new fact checking pipeline which removes the need for document retrieval and sentence scoring. I have previously worked on a classification problem where user queries were mapped to relevant services offered by a service provider. On very short or malformed sentences, this task was particularly challenging. To improve accuracy, I had to rely on subject-verb-object (SVO) extraction from the text before feeding it to the classifier. The paper discusses a similar problem, introducing some NLP optimizations to the text in the pre-training phase may help improve the accuracy of the language model. A general comment here is that it would be interesting to analyze the impact of LLMs on publisher revenue. LLMs are trained on data scrapped from the web, which has a direct cost for ad publishers (ad servers charge per impression) and people turning to LLMs for information (or in this case, fact checking), could have a much more drastic, though indirect, affect on the revenue for content publishers.